Luis Alejandro Morales-Marín

Education

2008–2013 **PhD.**, *University College London (UCL)*, London, United Kingdom, *Grade*.
Doctor of Philosophy in Physical Geography (Major in water resources)

2004–2007 **MSc.**, *Universidad Nacional de Colombia*, Bogotá, Colombia, *Grade*.
Master of Science in Water Resources Engineering

2002–2003 **Spe.**, *Escuela Colombiana de Ingeniería "Julio Garavito"*, Bogotá, Colombia, *Grade*.
Specialist in Water Resources and Environment

1997–2002 **BEng.**, *Escuela Colombiana de Ingeniería "Julio Garavito"*, Bogotá, Colombia, *Grade*.
Bachelor in Civil Engineering

Research Interest

Effects of climate and land-use changes on hydrological processes in river catchments.

Analysis of extreme hydrometeorological events: floods and droughts.

Large-scale hydrological modeling and land surface processes.

Hydraulic and hydrodynamic modelling in rivers and lakes.

Water quality and hydrological modelling in river catchments.

Professional Appointments

Aug 2022 - current **Assistant Professor**, *Department of Civil and Agriculture Engineering, Universidad Nacional de Colombia*, Bogota, Colombia.
Teaching and research in hydrological sciences.

Jan 2020 - Jul 2022 **Physical Scientist, Research**, *Numerical Environmental Prediction, National Prediction Development, Canadian Centre for Meteorological and Environmental Prediction, Environment and Climate Change Canada/Government of Canada*, Dorval, Canada.
Research in hydrological and hydrodynamical modelling and forecasting.

Jul 2018 - Dec 2019 **Research Associate**, *Global Institute for Water Security and Global Water Future Program, National Hydrology Research Centre, University of Saskatchewan*, Saskatoon, Canada.
Research associate in water quality and hydrological modelling in river catchments.

May 2013 - Jun 2018 **Postdoctoral Research Fellow**, *Global Institute for Water Security, National Hydrology Research Centre, University of Saskatchewan*, Saskatoon, Canada.
Postdoctoral Researcher in nutrient and sediment transport and large scale hydrological modeling.

April 2013 - Current **Invited Lecturer in River Science course**, *Global Institute for Water Security and School of Environment and Sustainability, University of Saskatchewan*, Saskatoon, Canada.
Implementation and evaluation of nutrient transport models in river catchments.

Hydraulic Laboratory (Building 409), office 305, Department of Civil and Agriculture Engineering, Universidad Nacional de Colombia, Bogotá –
☎ *+57 311 6459628* ▪ ✉ *lmoralesm@unal.edu.co* ▪ *lamhydro.github.io*

August 2008 - March 2012 **Graduate Assistant**, *University College London (UCL)*, London, United Kingdom.
IT assistant in the UNIX/LINUX laboratory. Support for UNIX/LINUX platforms and programming in Fortran, C and Matlab languages. Installation and support of specialized software in water resources such as: SWAN, SWAT and MIKE SHE. Preparation of explanatory manuals and documentation.

January 2006 - June 2006 **Lecturer in Hydraulic Structures**, *Universidad Distrital Francisco Jose de Caldas*, Bogotá, Colombia.
Lectures on design of water drainage systems and water supply structures. Lectures delivered to two groups of 30 students each; handouts and computational programs were provided to students.

January 2004 - March 2006 **Hydraulic Engineer**, *Aguazul Consulting S.A.*, Bogotá, Colombia.
Design of sewer and aqueduct networks for new housing developments in Bogotá, Colombia. Revision and design of urban drainage systems.

January 2004 - March 2006 **Supervision Engineer**, *Aguazul Consulting S.A.*, Bogotá, Colombia.
Technical and quality control of sewer and aqueduct networks in new housing developments in Bogotá. Revision of water drainage and waste-water system designs.

January 2003 - December 2003 **Assistant Engineer**, *Aguazul Consulting S.A.*, Bogotá D.C, Colombia.
Assistance during the supervision of hydraulic structure constructions such as channels and large sewer networks. Elaboration of technical reports for the Acueducto de Bogotá ESP S.A.

January 2002- December 2003 **Lecturer in Hydraulic Laboratories**, *Escuela Colombiana de Ingeniería "Julio Garavito"*, Bogotá, Colombia.
Lectures on pipe flow and open channel flow hydraulics laboratories. Lectures to 5 groups of 7 students each, including the elaboration of laboratory guidelines and revision of reports.

Awards and Achievements

May 2013 - Dec 2019 **Fellowship**, *Global Institute for Water Security*, Saskatoon, Canada, Postdoctoral Researcher in Water Resources Sciences.

September 2008 - March 2012 **Scholarship**, *University College London*, London, UK, Scholarship of the Department of Geography.

June 2005 - June 2006 **Representative**, *Universidad Nacional de Colombia*, Bogotá, Colombia.
Elected graduate student representative in the faculty of engineering.

Peer-reviewed Journal Papers

1. 2026 Chun, K., Octavianti, T., Bradford, L., Reeves, M., Nagheeby, M., Olusola, A., Howard, B., Ceperly, N., Castelli, G., Nkwasa, A., Budiyono, Y., Strickert, G., **Morales-Marín L. A.**, Jonson, S., Bezerra, R., Diele-Viegas, L. & Gopinath, D. (Accepted), Decolonising hydrology: Reflecting on positionalities for sustainable and just futures. *Environmental Science and Policy*.

2. Changhai Liu, Kyoko Ikeda, Francina Dominguez, Andreas F Prein, Roy M Rasmussen, Zhe Zhang, Lulin Xue, Richard B Neale, Kwok Pan Chun, Cenlin He, Rosimar Rios-Berrios, Michelle Simões Reboita, Yongjie Huang, Helber Barros Gomes, Marta Llopart, Jimy Dudhia, Yang Tian, Lucia Scaff, Adam C Varble, **Luis Morales-Marín**, Christoforus Bayu Risanto, Erin Dougherty, Jorge Llamocca Huamani, Daniel Veloso-Aguila, Kristen L Rasmussen, Aiguo Dai, Kelly M Núñez Ocasio, Zachary L Moon, Maria Laura Bettolli, Ye Mu, José Augusto Paixão Veiga, Everaldo Barreiros de Souza, Erick Claros, Leila Constanza Hernandez Rodriguez, Fei Chen, Zhe Feng, Maria Leidinice da Silva, Lluís Fita, Rocio Balmaceda-Huarte, Miguel Lagos-Zúñiga, Decker Guzman Zabalaga, K Santiago Hernández, Elizabeth Silvestre Espinoza, Franciano Scremin Puhales, José A Mantovani Jr, Marcelo Somos-Valenzuela, Juan Diego Giraldo-Osorio, Octavia Crompton, Omar V Müller, Timothy Schneider, Andrew J Newman, Pablo A Mendoza, Julio T Bacmeister, Patrick Callaghan, Wei Wang, Danqiong Dai, Cristian Martinez-Villalobos, Diego Urdiales-Flores, Luis Gustavo Gonçalves de Gonçalves, Ingrid Petry, Zhixiao Zhang, Tomásio E Januário, Angela Iza Wong, Santiago Nuñez Mejia, Víctor C Mayta, Flavio Justino, Germán Poveda, Peter J van Oevelen, Rosmeri Porfírio da Rocha, Álvaro Ossandón, Justin R Minder, Amanda Rehbein, Rong Fu, Marcelo Zamuriano, Liping Di, Ethan D Gutmann. (Accepted), Multi-Decadal Convection Permitting Dynamical Downscaling of Current and Future Climates over South America. *Climate Dynamics*.

3. **Morales-Marín, L. A.**, Rodríguez, E. & Jaramillo, F. (Accepted), Water resources challenges in Colombia: Hydrological science solutions for a sustainable future. *Hydrological Sciences Journal*.

4. 2025 **Morales-Marín, L. A.**, French, J., & Lindenschmidt, K. E. (2025), Exploratory Modelling of Wind-Generated Wave and Current Bottom Stress and Potential for Sediment Resuspension in Small Lakes. *Limnologica*, (114), 1-19.

5. Lucas, C., Gamazo, P., Crossa, M., Brum, E., Badagian, J., de los Santos, L., & **Morales-Marín, L. A.** (2025), Water Thermal Regime of River-Floodplain Systems in the Uruguay River Basin and Influence of Hydroclimatic Factors. *River Research and Applications*, (0), 1-19.

6. 2022 Akomeah E,**Morales-Marín L**, Lindenschmidt K-E and Hassanzadeh E. (2022), Evidence-based identification of integrated water quality systems. *Journal of Environmental Planning and Management*, 66 (7), 1431-1452.

7. 2021 **Morales-Marín L**, French J, Burningham H, Evans C and Burden A (2021), Simulating seasonal to multi-decadal variation in lake thermal response to meteorological forcing using the UCLAKE 1-dimensional model code. *Limnologica*, 88, 125874.

8. E.Akomeah, **L.A. Morales-Marín**, M. Carr, A. Sadeghian, K.E. Lindenschmidt (2021), The impacts of changing climate and streamflow on nutrient speciation in a large Prairie reservoir. *Journal of Environmental Management*, 288, 112262.

9. 2020 **Morales-Marín L**, Carr M, Sadeghian A and Lindenschmidt K-E (2020), Climate change effects on the thermal stratification of Lake Diefenbaker, a large multi-purpose reservoir *Canadian Water Resources Journal*, pp.1-16

10. Sadeghian A, **Morales-Marín L** and Lindenschmidt K-E (2020), Raw data and videos of historical and projected hydrodynamics and water quality in Lake Diefenbaker reservoir under climate change scenarios using the 2D CE-QUAL-W2 model. *Federated Research Data Repository*, https://doi.org/10.20383/101.0194

11. Rokaya P, **Morales-Marín L** and Lindenschmidt K-E (2020), A physically-based modelling framework for operational forecasting of river ice breakup. *Advances in Water Resources*, 103554

12. Carr M, Sadeghian A, Rinke K, Lindenschmidt K-E and **Morales-Marín L** (2020), Impacts of varying dam outflow elevations on water temperature, dissolved oxygen and nutrient distributions in a large prairie reservoir. *Environmental Engineering Science*.

2019

13. Sadeghian A, Lindenschmidt K-E, Carr M and **Morales-Marín L** (2019), Raw data and video outputs from a CE-QUAL-W2 hydrodynamic and nutrient model for different Lake Diefenbaker Reservoir withdrawal depth scenarios *Federated Research Data Repository*, http://dx.doi.org/10.20383/101.0134

14. Lindenschmidt K-E, Carr M, Sadeghian A and **Morales-Marín L** (2019), CE-QUAL-W2 model of dam outflow elevation impact on temperature, dissolved oxygen and nutrients in a reservoir. *Scientific Data*, 1-7.

15. **Morales-Marín L**, Rokaya, P., Sanyal, P.R. and Lindenschmidt, K.E., (2019), Changes in streamflow and water temperature affect fish habitat in the Athabasca River basin in the context of climate change. *Ecological Modelling*, 407, p.108718.

16. Rokaya P, **Morales-Marín L**, Bonsal, B, Wheater H and Lindenschmidt K-E, (2019), Climatic effects on ice phenology and ice-jam flooding of the Athabasca River in western Canada. *Hydrological Sciences Journal*, 64:11, 1265-1278.

17. **Morales-Marín L**, Sanyal, P.R., Kadowaki, H., Li, Z., Rokaya, P. and Lindenschmidt, K.E., (2019), A hydrological and water temperature modelling framework to simulate the timing of river freeze-up and ice-cover breakup in large-scale catchments. *Environmental Modelling & Software Journal*, 114, pp.49-63.

18. Hassanzadeh E , Strickert G, **Morales-Marín L**, Noble B, Baulch H, Shupena-Soulodre E and Lindenschmidt K-E, (2018) A framework for engaging stakeholders in water quality modeling and management: Application to the Qu'Appelle River Basin, Canada. *Journal of Environmental Management*, 231, 1117-1126

2018

19. **Morales-Marín L**, Wheater H and Lindenschmidt K-E, (2018) Potential changes of annual-averaged nutrient export in a large catchment under climate and land-use change scenarios. *Water*, 10(10), 1438.

20. **Morales-Marín L**, Wheater H and Lindenschmidt K-E (2018), Estimating sediment loadings in the South Saskatchewan River catchment. *Water Resources Management* 32: 769.

2017

21. **Morales-Marín L**, Wheater H and Lindenschmidt K-E (2017) Assessment of nutrient loadings of a large multipurpose prairie reservoir. *Journal of Hydrology* 550, 166-185.

22. **Morales-Marín L**, French J, Burningham H and Battarbee R W (2017), 3D hydrodynamic and sediment transport modelling to test the sediment focusing hypothesis in upland lakes. *Limnology and Oceanography*.

23. **Morales-Marín L**, French J and Burningham H and (2017), Application of a 3D ocean model to understand upland lake hydrodynamics and circulation. *Environmental Fluid Mechanics* 17: 1255.

2016 24. **Morales-Marín L**, K P Chun, Wheater H and Lindenschmidt K-E (2016), Trend analysis of nutrient loadings in a large prairie catchment. *Hydrological Sciences Journal*, 1-23.

2015 25. **Morales-Marín L**, Wheater H and Lindenschmidt K-E (2015), Assessing the transport of total phosphorus from a prairie river basin using SPARROW. *Hydrological Processes*, 29, 4144–4160.

In revision

López-Rativa, L. C., Das, A., Kowshal, A., **Morales-Marín L. A.** & Lindenschmidt K. E. (In revision), Integrating Long Short-Term Memory streamflow Forecasts into a Machine Learning Model for Ice Jam Flood Prediction. *Journal of Hydrology*.

Kwok P Chun, Thanti Octavianti, Georgia Papacharalampous, Hristos Tyralis, Samuel J Sutanto, Pavel Terskii, Paola Mazzoglio, Dario Treppiedi, Juan Rivera, Nilay Dogulu, Adeyemi Olusola, Bastien Dieppois, Moctar Dembélé, Simon Moulds, Cheng Li, **Luis Morales-Marín**, Neil Macdonald, Toundji Olivier Amoussou, Roland Yonaba, Salomon Obahoundje, Nicolas Massei, David M Hannah, Sivarama Krishna Reddy Chidepudi, Byman Hamududu (In revision), Unravelling compound risks of hydrological extremes in a changing climate: Typology, methods and futures. *WIREs Climate Change*.

Conference Proceedings, Presentations and Seminars

Conference proceedings

2020 1. Lindenschmidt, K.E., Akomeah E., **Morales-Marín L** and Hassanzadeh E., 2020, Nomvember. Posteriory indentification of a cold-region integrated catchment-river-lake water quality system. *25th IAHR International Symposium on Ice* At: Trondheim, Norway

2019 2. Rokaya, P., **Morales-Marín L** and Lindenschmidt, K.E., 2019, May. Towards Improved Real-time Forecasting of River Ice Breakup. *Proceedings of the 20th Workshop on the Hydraulics of Ice Covered Rivers* At: Ottawa, Canada

2018 3. Lindenschmidt, K.E., Akomeah, E., Baulch, H., Boyer, L., Davies, J.M., Hassanzadeh, E., **Marin, L.M.**, Strickert, G. and Wauchope, M., 2018, September. Interfacing Stakeholder Involvement into a Surface Water-Quality Modelling System for Water Management and Policy Development. *In International Conference on Urban Drainage Modelling* (pp. 312-316). Springer, Cham.

Conference presentations and seminars

1. Kwok Pan Chun, Ana Mijic, Luminita Danaila, Rosmeri Porfirio da Rocha, Thanti Octavianti, Liling Huang, Jesus Fernandez, Leonardo Aragao, Yasemin Ezber, Emir Toker, Andreas Hartmann, Yongping Wu, **Luis Morales-Marín**, C Bayu Risanto, Li Cheng, Lindsey McEwen 2025 *Weather Generator Based on Generative AI for Interdisciplinary Probabilistic Downscaling Using Convection-Permitting Model Outputs and Potential Utility in Equitable, Community-focused Climate Scenario-ing* European Geophysical Union, General Assembly Conference 2025. 27 April-2 May, 2025 in Vienna, Austria.

2. D. Durnford, M. Carrera, F. Dupont, D. Deacu, É. Gaborit, C. Garnaud, V. Fortin, S. Bélair, F. Lespinas, B. Bilodeau, D. Khedhaouiria, N. Gauthier, G. Roy, P. Matte, V. Vionnet, B. Badawy, X. Liu, B. Bekcic, Y. L. Shin, O. Champoux, M. Abahamowicz, S. Keita, **L. Morales-Marín**, J. Morin, Y. Hata, Y. Martinez 2021 *Hydrological prediction systems at Environment and Climate Change Canada* American Meteorological Society, 101st Annual Meeting 2021. Virtual.

3. Rokaya P, **Morales-Marín L** and Lindenschmidt, K.E., 2019 *A physically-based modelling framework for operational forecasting of river ice breakup floods* American Geophysical Union, Fall Meeting 2019. San Francisco, Dec., US.

4. P.F. Arboleda-Obando, E. Rodriguez, D. Princz, B. Davison, A. Ireson and **L.A. Morales**, 2019 *Using the MESH model to study climate and land use changes over a subcontinental tropical river basin in Colombia-South America* Canadian Geophysical Union (CGU), Joint Assembly 2019. Montreal, July 8 - 18, Canada.

5. **Morales-Marín L**, Sanyal P, Kadowaki H, Li Z, Rokaya P and Lindenschmidt K-E, 2018 *A hydrological and water temperature modelling framework to simulate the timing of river freeze-up and ice-cover breakup in large-scale catchments.* American Geophysical Union, Fall Meeting 2018. Washington D.C. Dec., US.

6. Li Z, Lindenschmidt K, Budhathoki S **Morales-Marín L** and Williams B, 2018 *Dynamic Contributing Drainage Area Mapping in the Prairie Pothole Region: Applying a Bayesian Deep Learning Technique to Multi-temporal Radarsat-2 Images.* American Geophysical Union, Fall Meeting 2018. Washington D.C. Dec., US.

7. Budhathoki S, **Morales-Marín L** and Lindenschmidt K, 2018 *Sediment Yield and Transport Model for Cold Region Catchments.* American Geophysical Union, Fall Meeting 2018. Washington D.C. Dec., US.

8. Strickert G, Hassanzadeh E, Noble B, Baulch H, **Morales-Marín L** and Lindenschmidt K, 2017 *Putting people into water quality modelling.* American Geophysical Union, Fall Meeting 2017. New Orleans Dec., US.

9. **Morales-Marín L**, Wheater H, Lindenschmidt K and Yassin F 2017 *A new solute transport model for large scale cold region catchments: A theoretical framework.* Canadian Geophysical Union (CGU), Joint Assembly 2017. Vancouver May. 28 - 31, Canada.

10. Rokaya P, **Morales-Marín L**, Wheater H and Lindenschmidt K-E, 2017 *Hydroclimatic Variability and implications for ice-jam flooding in the Athabasca River Basin in western Canada* Canadian Geophysical Union (CGU), Joint Assembly 2017. Vancouver May. 28 - 31, Canada.

11. **Morales-Marín L**, Wheater H and Lindenschmidt K, 2016 *Impacts of climate and land use changes on regional nutrient export in the South Saskatchewan River catchment.* American Geophysical Union, Fall Meeting 2016. San Francisco Dec., US.

12. Javid H, Davison B, Princz D, Rokaya P, Sapriza-Azuri G, Wheater H, **Morales-Marín L** and Lindenschmidt K 2015 *Assessing the Impacts of Reservoir Regulations and Climate Variability on the Peace River Runoff and Peace-Athabasca-Delta Using a Distributed Hydrological Model.* American Geophysical Union (AGU), Fall Meeting 2015. San Francisco Dec., US.

13. **Morales-Marín L**, Wheater H and Lindenschmidt K, 2015 *Regional nutrient export modeling of the South Saskatchewan River catchment using SPARROW.* Canadian Geophysical Union (CGU), Joint Assembly 2015. Montreal May. 3 - 7, Canada.

14. **Morales-Marín L**, Wheater H and Lindenschmidt K, 2014 *Regional total phosphorus export modeling of the Red Deer River catchment using SPARROW.* Canadian Geophysical Union (CGU), General Meeting 2014. Banff May. 4 - 7, Canada.

15. **Morales-Marín L**, French J and Burningham H, 2012 *Hydrodynamic modelling of small upland lakes under strong wind forcing.* European Geophysical Union(EGU), General Assembly 2012. Vienna Apr. 22 - 27, Austria.

16. **Morales-Marín L**, French J, Burningham H, Evans C and Battarbee R, 2011 *Upland lake hydrodynamics and its interaction with meteorological forcings.* UCL Graduate School Research Poster Competition. London April, UK.

17. **Morales-Marín L**, French J, Burningham H, Evans C and Battarbee R, 2010 *Wind forcing of upland lake hydrodynamics: implementation and validation of a 3D numerical model.* American Geophysical Union, Fall Meeting 2010. San Francisco Dec., US.

Scholarly Thesis

PhD thesis

Title *Numerical modelling of hydrodynamics and sedimentation in upland lakes: a test of sediment focusing hypothesis.*

Supervisors Prof. Jon French and Dr. Helene Burningham

Description Study of the dynamics of upland lakes using computational models to understand the effects of climate forcing on hydrodynamics and sedimentation processes. Extensive field work was undertaken with the aim of collecting information to calibrate and validate the computational models.

Master thesis

Title *Implementation and evaluation of a mathematical model to simulate the routing of a hydrograph in the Dique Channel at the Magdalena River.*

Supervisors Dr. Luis Alejandro Camacho

Description Development of a computational model using finite difference methods to resolve the 2D Navier-Stokes equations. The model was calibrated and validated against hydrographs recorded at hydrometric station along the Magdalena River (Colombia) nearby the Dique Channel.

Specialization thesis

Title *Numerical methods to solve the Saint-Venant equations in super-critical regime. A comparison against experimental data.*

Supervisors Dr. German Ricardo Santos

Description Development of algorithms using finite difference methods to resolve the Saint-Venant equations in super-critical regime. The algorithms were calibrated against discharge and water level measurements taken in a prismatic flume equipped with automatic sensors.

Teaching Activities

Graduate courses

2020482 **Graduate Research Seminar I**, *Universidad Nacional de Colombia*, 2024 - current.

2020483 **Graduate Research Seminar II**, *Universidad Nacional de Colombia*, 2024 - current.

2025828 **Stochastic Methods in Water Resources**, *Universidad Nacional de Colombia*, 2024 - current.

ENVS 824.3 **River Science**, *Saskatchewan University*, Fall 2015 - Fall 2016.

Undergraduate courses

2015966 **Fluids Mechanics**, *Universidad Nacional de Colombia*, 2022 - current.

2015961 **Basic Hydraulics**, *Universidad Nacional de Colombia*, 2024 - current.

2015954 **Hydraulic Structures**, *Universidad Nacional de Colombia*, 2024 - current.

Grants

Oct-2025 to Oct-2026 **Hydrological Forecasting System for the Analysis of Flood Vulnerability and Risk in the Guatiquía River Basin, Meta, Colombia**, *Universidad Nacional de Colombia*, Bogotá, Colombia, National Call for Solidarity Extension Projects 2025. US $ 13.000. Principal investigator.

Jun-2025 to Jun-2026 **Validation of Water Levels in the San Rafael and Chuza Reservoirs of the Chingaza System Using the Surface Water and Ocean Topography (SWOT) Satellite**, *Universidad Nacional de Colombia*, Bogotá, Colombia, Faculty of Engineering Research Projects Call 2025. US $ 13.000. Principal investigator.

Apr-2025 to Apr-2026 **System for the Analysis and Early Warning of Meteorological and Hydrological Droughts for Bogotá and the River Basins and Reservoirs that Supply Water to the City**, *Universidad Nacional de Colombia*, Bogotá, Colombia, Agencia Distrital para la Educación Superior, la Ciencia y la Tecnología (ATENEA) Call. US $ 53.000. Co-principal investigator.

2016 **A water quality modeling system of the Qu'Appelle River catchment for longterm water management policy development**, *Global Institute for Water Security*, Saskatoon, Canada, Support in the preparation of the project proposal. CAN $ 309.578.

2011 **Field Work**, *University College London*, London, UK, Support for field work carried out in the north of Wales, UK .US $ 1.000.

2010 **Conferences**, *University College London*, London, UK, Maximum amount to attend to the 2010 American Geophysical Union Fall Meeting .US $ 1.800.

2006 **Research project**, *Universidad Nacional de Colombia*, Bogota, Colombia.
Grant to support master's research projects. US $ 5.000. Co-principal investigator

Professional and Public Services Activities

Professional association membership

2023 - Current **IAHR**, *International Association for Hydro-Environment Engineering and Research*, Worldwide.

2010 - Current **AGU**, *American Geophysical Union*, United States.

Peer reviewer for journals

2017-continued **Journal of Geophysical Research:Bioscience**.

2017-continued **Journal of Water and Land Development**.

2017-continued **Science of the Total Environment**.

2016-continued **Water**.

2016-continued **Water Resources Management**.

2015-continued **Journal of Hydrology**.

2012-continued **Limnology and Oceanography**.

Organization of scientific meeting sessions

American Geophysical Union (AGU) **AGU Fall meeting 2016**, *New Frontiers in Water Resources: Achieving Water Resource Security in Times of Climate Change, Urbanization, and Agricultural Expansion*, Convener.

Administrative service at universities

2024 - Current **Master's Program Coordinator**, *Master in Water Resources Engineering*, Universidad Nacional de Colombia.

Mentoring and Advising Activities

Undergraduate student supervision and co-supervision

Daniel Rivera, Department of Civil and Agriculture, Universidad Nacional de Colombia, *Degree sought: BEng*, Start year: 2022 - current, Supervisor.

Alex Mateus, Department of Civil and Agriculture, Universidad Nacional de Colombia, *Degree sought: BEng*, Start year: 2022 - current, Supervisor.

Jorge Méndez, Department of Civil and Agriculture, Universidad Nacional de Colombia, *Degree sought: BEng*, Start year: 2021 - current, Supervisor.

Graduate student supervision and co-supervision

Sebastian Norato, Department of Civil and Agriculture, Universidad Nacional de Colombia, *Degree sought: MSc*, Start year: 2026 - current, Supervisor.

Nicolas Muñoz, Department of Civil and Agriculture, Universidad Nacional de Colombia, *Degree sought: MSc*, Start year: 2026 - current, Supervisor.

Daniela Ortega, Department of Civil and Agriculture, Universidad Nacional de Colombia, *Degree sought: MSc*, Start year: 2026 - current, Supervisor.

Mariana Ulchur, Department of Civil and Agriculture, Universidad Nacional de Colombia, *Degree sought: MSc*, Start year: 2026 - current, Supervisor.

Daniel Salas, Department of Civil and Agriculture, Universidad Nacional de Colombia, *Degree sought: MSc*, Start year: 2026 - current, Supervisor.

Edgar Virguez, Department of Civil and Agriculture, Universidad Nacional de Colombia, *Degree sought: MSc*, Start year: 2026 - current, Supervisor.

Andres Pinto, Department of Civil and Agriculture, Universidad Nacional de Colombia, *Degree sought: MSc*, Start year: 2026 - current, Supervisor.

Jose Cortéz, Department of Civil and Agriculture, Universidad Nacional de Colombia, *Degree sought: MSc*, Start year: 2026 - current, Supervisor.

Jose Alfonso, Department of Civil and Agriculture, Universidad Nacional de Colombia, *Degree sought: MSc*, Start year: 2025 - current, Supervisor.

Lucia Salazar, Department of Civil and Agriculture, Universidad Nacional de Colombia, *Degree sought: MSc*, Start year: 2025 - current, Supervisor.

Luis López, Department of Civil and Agriculture, Universidad Nacional de Colombia, *Degree sought: MSc*, Start year: 2024 - current, Supervisor.

Daniel Barreto, Department of Civil and Agriculture, Universidad Nacional de Colombia, *Degree sought: MSc*, Start year: 2024 - current, Supervisor.

Jorge Leal, Department of Civil and Agriculture, Universidad Nacional de Colombia, *Degree sought: MSc*, Start year: 2024 - current, Supervisor.

Thomas Aranda, Department of Civil and Agriculture, Universidad Nacional de Colombia, *Degree sought: MSc*, Start year: 2024 - current, Co-supervisor.

Oscar Camargo, Department of Civil and Agriculture, Universidad Nacional de Colombia, *Degree sought: MSc*, Start year: 2024 - current, Supervisor.

Julio Gonzales, Department of Civil and Agriculture, Universidad Nacional de Colombia, *Degree sought: MSc*, Start year: 2023 - current, Supervisor.

Edwin Amaya, Department of Civil and Agriculture, Universidad Nacional de Colombia, *Degree sought: MSc*, Start year: 2023 - current, Supervisor.

Jonathan Romero, Department of Civil and Agriculture, Universidad Nacional de Colombia, *Degree sought: MSc*, Start year: 2021 - End year: 2024, Co-supervisor.

Eric Akomeah, Global Institute for Water Security, University of Saskatchewan, *Degree sought: PhD*, Start year: 2016 - End year: 2021, Co-supervisor.

Graduate board exam committee membership

Laura Constanza Torres Torres, Civil and Agriculture Engineering Department, Universidad Nacional de Colombia, *Awarded degree: M.Sc. in Water Resources Engineering*, 2025.

Lina Fernanda Delgado Peña, Civil and Agriculture Engineering Department, Universidad Nacional de Colombia, *Awarded degree: M.Sc. in Water Resources Engineering*, 2024.

Leonardo Albornoz Villarraga, Civil and Agriculture Engineering Department, Universidad Nacional de Colombia, *Awarded degree: M.Sc. in Water Resources Engineering*, 2024.

David Andrés Zamora Ávila, Civil and Agriculture Engineering Department, Universidad Nacional de Colombia, *Doctoral Thesis Project and Defense Examiner*, 2018.

Jaime Alberto Parra Plazas, Electrical and Electronics Engineering Department, Universidad Distrital "Francisco José Caldas", *Doctoral Thesis Project and Defense Examiner*, 2017.

Alvaro Enrique Ortiz Dávila, Computing Engineering Department, Universidad Distrital "Francisco José Caldas", *Doctoral Thesis Project and Defense Examiner*, 2017.

Lina Sofia Amaya Toro, Civil and Agriculture Engineering Department, Universidad Nacional de Colombia, *Awarded degree: M.Sc. in Water Resources Engineering*, 2016.

Languages

Spanish	**Good**	*Reading, writing, listening, speaking*
English	**Good**	*Reading, writing, listening, speaking*
French	**Basic**	*Reading, writing, listening, speaking*

Computer Skills

Programming language	Fortran77/90, C/C++	Scripting language	Shell, Python
Operative Systems	UNIX/LINUX, Windows	Numerical Software	Matlab, R
Specialized software	ArcGIS, HEC-RAS, WASP, CE-QUAL-W2, POM/FVCOM, SMS, VIC, SWAT, SWAN	Word processor	LaTeX, Microsoft Office

Laboratory and Field Work Skills

Hydraulic laboratory	Use of prismatic flumes, pipe systems and pumping systems.	Field work	Use of ADCP Acoustic Doppler Current Profiler), pressure and temperature sensors, and meteorological stations and acoustic echoe-sounder for bathymetry surveys.

Sedimentology laboratory	Use of sizer laser to know grain distribution in sediment samples.

Interests

Professional	Computational hydrology and hydraulics	Software	Linux and high performance computing
Health	Bicycling, Swimming	Languages	English, French